

AHMAD SOBOH

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EDUCATION

University of Ottawa – *Bachelor of Applied Science (Co-op), Software Engineering*

Dec 2027

- GPA: 3.92 / 4.0 (Dean's Honour List)
- Relevant Coursework: Data Structures and Algorithms, Network Engineering, Operating Systems, Computer Architecture, Databases (SQL), Programming Paradigms, Discrete Structures, Introduction to Software Engineering, Digital Systems, Software Construction

EXPERIENCE

Nokia

Software Engineer Intern

Sept 2026 – Dec 2026

Ottawa, Canada

- Enhanced TDM over TCP/IP capabilities across Nokia SR/SAR routers by developing Tcl software for large-scale network infrastructure
- Expanded automated routing validation across 20+ test scenarios by developing Python/Tcl tests for Nokia SAR routers using PyTest
- Reduced manual router config by 90% by building Python and Bash automation for router setup, config generation, and test execution
- Diagnosed 10+ networking failures by tracing packet flows, logs, and system state across router software, and delivering fixes

Ciena

ASIC Design and Verification Hardware Intern

Jan 2025 – Aug 2025

Ottawa, Canada

- Closed 200+ functional coverage gaps across 2 critical subsystems by validating ASIC functionality for a 1600G optical system
- Achieved 100% functional coverage across ASIC signals by developing C++ and SystemVerilog/UVM coverage models
- Eliminated 100+ hours of redundant verification effort by designing a reusable UVM scoreboard architecture for SERDES
- Increased functional coverage by 20% by developing C++ FEC verification tests and integrating them with coverage models
- Reduced failure debug time by 17% by engineering a Python regression pipeline that processed 10,000+ nightly test results

Ford

Vehicle Software Engineer Intern

Sept 2024 – Dec 2024

Ottawa, Canada

- Engineered Python test automation for embedded vehicle software, executing integration tests and triaging 1,000+ production defects
- Automated firmware deployment across 50+ vehicle ECUs with Python and Jenkins CI/CD, accelerating embedded software validation
- Resolved 20+ critical ECU software failures by developing Python log-analysis tools, reproducing vehicle issues, and parsing ECU logs
- Test-drove vehicles running new software releases, reproduced on-road issues, and used CAN FD cables to test specific ECU's

Ford

Vehicle Software Engineer Intern

Jan 2024 – Apr 2024

Ottawa, Canada

- Debugged vehicle software issues by reproducing failures, tracing ECU logs, and validating fixes through automated integration testing
- Investigated 15+ vehicle software failures with cross-functional teams, identified root causes, and delivered fixes across embedded systems
- Built a defect analytics platform using React, Node.js, and Jira API to analyze 1,000+ defects and cut recurring issues by 35%

Masv

Software Engineer Intern

May 2023 – Aug 2023

Ottawa, Canada

- Cut customer support ticket resolution time by 8 minutes by building a Vue, Node.js, PostgreSQL admin platform for user-data retrieval
- Reduced deployment failures by 50% by enforcing automated pull-request testing and improving CI/CD pipelines
- Reduced backend query latency by 500ms by engineering APIs and optimizing complex SQL queries for large file-transfer datasets
- Improved build stability across dev and prod by containerizing application services with Docker and deploying them on AWS EC2

PROJECTS

HeartAI – AI-powered Human Heart Defect Scanner | *Python, PyTorch, MONAI, 3D Slicer*

Sept 2026

- Built an AI-powered system that analyzes cardiac CT scans to help doctors identify heart defects and reconstruct the 3D heart models
- Used a PyTorch/MONAI 3D segmentation model to detect defects in cardiac chambers and major vessels from medical images
- Engineered a GPU-accelerated Python pipeline for CT preprocessing, AI inference, 3D heart reconstruction, and anatomical measurements
- Developed an interactive React/Three.js interface to visualize reconstructed hearts and investigate suspected abnormalities in 3D

LettuceEat – 1st Place Hackathon Winner (uOttHack 2026) | *GeminiAPI*

Jan 2026

- Developed a full-stack web application that uses AI to browse restaurants, compare options, and coordinate group meal plans

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, Bash, SystemVerilog (UVM)

Frameworks/Libraries: React, Next.js, Node.js, Express.js, Vue.js, Spring Boot, Django, Pytest, JUnit, NumPy, Ruby on Rails, jQuery

Cloud & Infrastructure/Developer Tools: AWS, Docker, Kubernetes, Linux, QNX, Jenkins, Git, PostgreSQL, Jira, Confluence, Firebase, Supabase, Android Studio, macOS, Claude, Cursor, Codex

Networking: TCP/IP, Routing, Packet Forwarding, Network Protocols, CAN Bus